# SOFTWARE REQUIREMENT SPECIFICATION (SRS)

**Security Incident Mapping (SIM)**

**Course: SEN202**

**Project Title: Security Incident Mapping (SIM)**

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# 1. Introduction

## 1.1 Purpose

The Security Incident Mapping (SIM) system is a web-based application developed to improve security within the campus community. It enables students, staff, and landlords to report security incidents quickly while providing administrators with a centralized dashboard to monitor and manage reported incidents. The aim is to encourage prompt reporting, improve emergency response, and support informed decision-making for campus security management.

## 1.2 Scope

The system allows members of the campus community to report security incidents through a simple web interface. It stores incident reports, enables administrators to monitor and manage reports, and provides basic statistics on reported incidents. The application is designed as a prototype for academic purposes and demonstrates the fundamental features of a digital crime reporting platform.

## 1.3 Objectives

**The objectives of the Security Incident Mapping system are:**

- **To provide an easy and fast method for reporting security incidents.**
- **To reduce delays in reporting crimes within the campus environment.**
- **To assist administrators in monitoring reported incidents.**
- **To improve campus security through digital reporting.**
- **To replace manual reporting with an efficient electronic system.**

## 2. Overall Description

### 2.1 Product Perspective

**The Security Incident Mapping (SIM) system is a standalone web application developed using HTML, CSS, and JavaScript. The application stores information using the browser's Local Storage and does not require a server or online database. It serves as a prototype that demonstrates the basic operations of a security incident reporting system.**

## 2.2 Product Functions

**The system provides the following functions:**

- **Submit new security incident reports.**
- **Store reports locally in the browser.**
- **Display all submitted reports.**
- **Search reports by category or location.**
- **Filter reports based on incident category.**
- **Update the status of reported incidents.**
- **Delete reports.**
- **Display dashboard statistics.**

## 2.3 User Classes

**Student**

**Students can:**

- **Report security incidents.**
- **Upload supporting evidence.**
- **View safety information.**
- **Track submitted reports.**

**Staff**

**Staff members can:**

- **Report security incidents.**
- **Upload supporting evidence.**
- **Track submitted reports.**

**Landlord**

**Landlords can:**

- **Report incidents occurring around student residences.**
- **Upload supporting evidence.**
- **Track submitted reports.**

**Administrator**

**The administrator is responsible for:**

- **Logging into the system.**
- **Viewing all incident reports.**
- **Searching reports.**
- **Filtering reports.**
- **Updating report status.**
- **Deleting reports.**
- **Viewing dashboard statistics.**

## 3. Functional Requirements

### FR1 – Home Page

**The system shall display a homepage containing:**

- **Navigation menu**
- **About section**
- **Features section**
- **Statistics section**
- **Report Incident button**

### FR2 – Incident Reporting

**The system shall allow users to submit incident reports containing:**

- **Full Name**
- **Anonymous reporting option**
- **Email Address**
- **Phone Number**
- **Incident Category**
- **Date**
- **Time**

- Location
- Severity Level
- Description of the incident

**FR3 – Data Storage**

**The system shall store submitted reports using the browser's Local Storage.**

**FR4 – Dashboard**

**The administrator dashboard shall display:**

- **Total reports**
- **Pending reports**
- **Resolved reports**
- **Theft cases**
- **Recent incident records**

**FR5 – Search Function**

**The administrator shall be able to search reports using:**

- **Incident location**
- **Incident category**

**FR6 – Filter Function**

The administrator shall be able to filter reports according to the selected incident category.

FR7 – Report Status

The administrator shall be able to update the status of reports from Pending to Resolved.

FR8 – Delete Report

The administrator shall be able to permanently delete incident reports.

4. Non-Functional Requirements

Performance

- The system should load within three seconds under normal conditions.
- Dashboard updates should occur immediately after a report is submitted.

Reliability

- Submitted reports should remain available until deleted by the administrator.
- Required fields must be validated before submission.

Security

- **User input should be validated.**
- **The administrator dashboard should only be accessible after authentication in future versions.**

**Usability**

**The system should:**

- **Be simple to learn.**
- **Provide a user-friendly interface.**
- **Support desktop and mobile devices.**

**Maintainability**

**The source code should be modular, readable, and easy to modify.**

**Portability**

**The application should operate correctly on modern web browsers including:**

- **Google Chrome**
- **Microsoft Edge**
- **Mozilla Firefox**
- **Opera**

## 5. System Design

## Input

**The system accepts the following information:**

- **Full Name**
- **Email Address**
- **Phone Number**
- **Incident Category**
- **Date**
- **Time**
- **Location**
- **Severity Level**
- **Description of the incident**

## Processing

**The system performs the following operations:**

1. **The user fills in the incident report form.**
2. **JavaScript validates the input data.**
3. **The report is saved to Local Storage.**
4. **The dashboard retrieves all stored reports.**
5. **Statistics are automatically generated.**

6. The administrator manages the submitted reports.

**Output**

**The system produces:**

- **Confirmation of successful report submission.**
- **Dashboard containing all incident reports.**
- **Statistics of reported incidents.**
- **Search and filter results.**
- **Updated report status.**
- **Deleted report records.**

## 6. Object Listing

| Object | Description |
|---|---|
| User | A student, staff member, or landlord who submits an incident report. |
| Administrator | The person responsible for managing reports. |
| Incident Report | A record containing details of a reported incident. |
| Dashboard | Displays reports and system statistics. |

| Object | Description |
|---|---|
| Report Form | Collects incident information from users. |
| Local Storage | Stores reports in the browser. |
| Search Module | Searches reports based on keywords. |
| Filter Module | Filters reports according to category. |
| Statistics Module | Calculates and displays report summaries. |

## 7. Future Improvements

Future versions of the system may include:

- GPS-based incident location tracking.
- Google Maps integration.
- SMS notifications.
- Email notifications.
- AI-powered crime prediction.
- Secure user authentication.
- Cloud database integration.
- Mobile application support.
- Image recognition for uploaded evidence.

- Integration with police and emergency response agencies.

## 8. Conclusion

The Security Incident Mapping (SIM) system provides an efficient and user-friendly approach to reporting and managing security incidents within a campus community. By allowing students, staff, and landlords to submit reports electronically and enabling administrators to monitor these reports through a centralized dashboard, the system contributes to faster reporting, improved communication, and better security management. Although this prototype uses Local Storage for demonstration purposes, it establishes a solid foundation for future enhancements such as cloud storage, real-time location mapping, user authentication, and integration with emergency response services.

### Functional Requirements Summary

| Requirement ID | Description |
|---|---|
| FR1 | Display the Home Page |
| FR2 | Submit Incident Reports |

| Requirement ID | Description |
|---|---|
| FR3 | Store Reports in Local Storage |
| FR4 | Display the Dashboard |
| FR5 | Search Reports |
| FR6 | Filter Reports |
| FR7 | Update Report Status |
| FR8 | Delete Reports |

**Non-Functional Requirements Summary**

| Requirement ID | Description |
|---|---|
| NFR1 | Fast system response time |
| NFR2 | Reliable report storage |
| NFR3 | Input validation and basic security |
| NFR4 | User-friendly interface |
| NFR5 | Responsive design |
| NFR6 | Easy maintenance |
| NFR7 | Cross-browser compatibility |